# Hugo to Apollo Sync API - Design

## Overview

The **Sync Domain API** provides endpoints that sends messages published by HUGO/BOSS to Azure Service Bus that are synched to Apollo

### Key Entities

The following are the core business entities related to sync within the Domain API context.

- **Sync Batch**
  - Entity related to sending Sync Messages as a batch
  - Send messages to requested Queue
- **CRM Sync Batch**
  - Entity related to the CRM Sync Batch Message
  - Send message to requested CRM

**Technical Product Owner(s):** Anbarasan Rajangam

**Stakeholders:**

The key stakeholders are listed below, for the full list of project stakeholders refer [Project Stakeholders - DotNet Team - Confluence]

| Upstream App name | Technical Owner |
|---|---|
| Titan DataStream | Damian Wensing |

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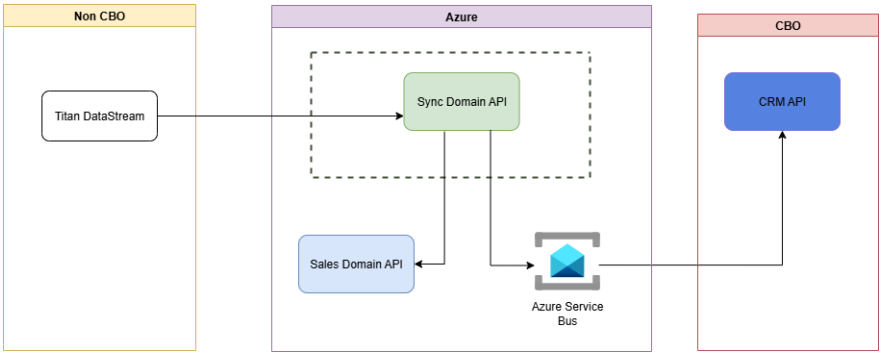
## High-Level Integration Diagram

Below is the high-level interaction diagram showing key upstream applications and other external/internal systems

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## Domain Model

### Non-Restful Model

```
{
    "syncBatch": {
        "partitionKey": "long",
        "messages": [{
            "messageId": "string",
            "transactionId": "string",
            "syncMethod": "string", //enum: Create, Update, Delete
            "entityType": "string", //enum: Person,PersonMembership,PersonContact,ExternalCard,MedicareCard,OverseasPerson,PersonStudySummary,PersonStudyDetails,QuoteStatus,PersonRole,OrderFulfilmentStatus,Order,SalesTransaction,Membership,MembershipPackage,CompoundPersonMemberships, CompoundPolicyStatusHistory, // Compound Policy Status History PolicyStatusHistory,ChangeOfCover,PersonCommunicationDetails,PersonCommunicationPreferences, //Communication Preferences update compound MessageCommPreference,PersonFlags,EntityFlag,MembershipFlags,PersonStudy,CorporateGroupEntitlement,CorporateGroupPolicyEntitlement
            "streamId": "string",
            "entityId": "long",
            "timestamp": "string",
            "body": "JToken" //json
        }]
    }


    "crmSyncBatch": {
```

```
        "syncFailedMessages": "bool",
        "messages": [{
            "messageId": "string",
            "transactionId": "string",
            "syncMethod": "string", //enum: Create, Update, Delete
            "entityType": "string", //enum: Person,PersonMembership,PersonContact,ExternalCard,MedicareCard,OverseasPerson,PersonStudySummary,PersonStudyDetails,QuoteStatus,PersonRole,OrderFulfilmentStatus,Order,SalesTransaction,Membership,MembershipPackage,CompoundPersonMemberships, CompoundPolicyStatusHistory, // Compound Policy Status History PolicyStatusHistory,ChangeOfCover,PersonCommunicationDetails,PersonCommunicationPreferences, //Communication Preferences update compound MessageCommPreference,PersonFlags,EntityFlag,MembershipFlags,PersonStudy,CorporateGroupEntitlement,CorporateGroupPolicyEntitlement
            "streamId": "string",
            "entityId": "long",
            "timestamp": "string",
            "body": "JToken"  //json
        }]
    }
}
```

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#### Upstream and Downstream Systems

- **Upstream systems:** The following systems shall access Sync Domain API
  - Titan Datastream Service
- **Downstream systems:** The following systems shall be accessed by Sync Domain API
  - Azure Service Bus
  - CRM

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#### Caching

Caching Component Details

[Caching Design - DotNet Team - Confluence]

Integration Approach

How to integrate HI.Library.Cache component with domain API

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### Authentication

Authentication for the Membership Domain API is implemented using **Bearer Token Authentication** (OAuth 2.0 + JWT).

- **Client Applications** obtain an access token from **Microsoft Entra** using either:
  - **User Credentials** (Public Clients), or
  - **Managed Identity** (Confidential Clients).
- The access token **must** include the required permissions as **scopes** or **roles**.
- For every request to the Sync Domain API, the client application must send the access token as a **Bearer token** in the `Authorization` header:

```
Authorization: Bearer <access_token>
```

- The Bearer token is validated, and access to the requested endpoint is granted accordingly.

#### Additional References

- **Thick Client to Domain API (Cyclops to Domain API)**
  Thick Client to Domain API (Cyclops to Domain API)- Authentication and Authorization Flow
- **API to API Flow (BFF to Domain API)**
  [Authentication and Authorization Flow – DotNet Team - Confluence]

#### Token Validation Logic

The JWT is validated at the **API Management (APIM)** level for every request. Validation checks include:

- Valid Issuer
- Valid Audience
- Token Signature
- Token Lifetime

If any validation fails, the request is rejected with a **401 Unauthorized** response.

> Note: Similar token validation functionality exists within the `HI.Library.Authorization` package and can be enabled or disabled via configuration. Currently, this package's token validation is bypassed because tokens are validated at the APIM level.

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### Authorisation

#### Authorisation Overview

Authorisation is performed based on the **scopes** or **role claims** present in the JWT bearer token of each request. The `HI.Library.Authorization` package is utilized by the Sync Domain API to implement the authorisation logic.

#### How is Authorisation Implemented?

- All applicable permissions such as `SyncDomainAPI.Create`, `SyncDomainAPI.Read`, etc., will be initialized for authorisation purposes.
- Endpoints requiring authorisation will be annotated with the `[AuthorizePermission]` custom attribute, specifying the required permission as an argument.
  **Example:**
  ```
  [AuthorizePermission("SyncDomainAPI.Read")]
  ```
  - This would typically be used on an HTTP GET method.
    Please refer to the complete list of permissions in the Hugo to Apollo Sync API - Design | Mapping of Users/Groups to Roles section.
  - When a request is made to an endpoint, the authorisation handlers validate that the required permissions (specified in the `[AuthorizePermission]` attribute) exist within the `scope` or `roles` claim of the JWT.
  - If the required claims are present and valid, access to the endpoint is granted. Otherwise, the request is rejected with a **401 Unauthorized** response.

#### Mapping of Users/Groups to Roles

> **Note:** This section applies to **Public Client Applications** only.

- For **Cyclops**, users will be added to the Azure Entra group:
  `RES-CLD-Cyclops-Hugo-dev-access`
  This group is associated with the App Registration during User Onboarding.
- Similarly, other public client applications (involving user interaction) can have their own groups, which can be assigned to their respective App Registrations.

#### Per-Endpoint Access Rules

- Each endpoint **must** use the `[AuthorizePermission]` custom attribute provided by the `HI.Library.Authorization` package.
- The following permissions apply to the Sync API Endpoints (list to be provided in the detailed API design documentation).

| Permission | Applicable HTTP Verb |
|---|---|
| `SyncDomainAPI.Create` | `POST` |

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### Logging

The **Logging component** provides a centralized logging solution for Hugo Domain APIs. It is implemented as a separate class library, enabling seamless integration across microservices as a reusable NuGet package.

- **Package Name:** `HI.Library.Logging`
- **Logging Design Documentation:**
  Logging - Design

#### Architecture & Implementation

- Logs are sent to **Azure Application Insights** for centralized monitoring and analysis.

- Uses the `ILogger` interface from **Microsoft.Extensions.Logging** to capture logs across Domain APIs.
- **OpenTelemetry** is leveraged to collect and export logs to Azure Application Insights via the `Azure.Monitor.OpenTelemetry.AspNetCore` package.

#### Key Logged Fields

| Field Name | Description | Field Type |
|---|---|---|
| Timestamp | Date and time when the log entry was created. | Telemetry |
| Message | Main content of the log describing the event or issue. | Telemetry |
| Severity Level | Severity of the log (Info(1), Warning(2), Error(3), Critical(4)). | Telemetry |
| Operation ID | Unique identifier for the API request to correlate logs across services. | Telemetry |
| Operation Parent Id | ID of the parent operation to correlate nested operations (e.g., request & dependencies). | Telemetry |
| Cloud Role Name | Role name of the cloud service where telemetry data originated. | Telemetry |
| Method | HTTP method used in the request (GET, POST, PUT, DELETE). | Telemetry / Custom |
| Path | URL path of the request indicating the accessed endpoint. | Telemetry / Custom |
| Trace Id | Unique identifier for a single trace. | Telemetry |
| Item Type | Type of telemetry item (request, exception, trace). | Telemetry |
| Session ID | Identifier for a user session tracking user actions over a timeframe. | Custom |
| Customer ID | Unique identifier for a customer. | Custom |
| CorrelationID | Custom correlation ID to trace and correlate requests/responses across distributed services. | Custom |

#### Log Levels and Retention

| Log Level | Description |
|---|---|
| Information | Informational messages about normal application operation. |
| Warning | Potential issues or unexpected behaviors that do not disrupt application flow. |
| Error | Application errors or exceptions preventing specific operations from succeeding. |
| Critical | Severe errors causing application or critical component failures. |

#### Retention Policy

- Application Insights retains log data for a configurable retention period (90 days).

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## Exception and Error Handling

#### Global Exception Handling Pattern

Exceptions are handled using the **Global Exception Handling Middleware** provided by the `HI.Library.ExceptionHandling` package.

#### Mapping Exception Types to HTTP Status Codes

| Exception Type | HTTP Status Code | Description |
|---|---|---|

| | | |
|---|---|---|
| Bad Request | 400 Bad Request | The server could not understand the request due to invalid syntax. |
| Unauthorized | 401 Unauthorized | Authentication is required and has failed or has not yet been provided. |
| Forbidden | 403 Forbidden | The client does not have access rights to the content. |
| NotFound | 404 Not Found | The server can't find the requested resource. |
| RequestTimeout | 408 Request Timeout | The server timed out waiting for the request. |
| Conflict | 409 Conflict | The request conflicts with the current state of the server. |
| PreconditionFailed | 412 Precondition Failed | The server does not meet one of the preconditions in the request headers. |
| Unprocessable Request | 422 Unprocessable Entity | The request was well-formed but unable to be followed due to semantic errors. |
| InternalServerError | 500 Internal Server Error | The server has encountered a situation it doesn't know how to handle. |
| NotImplemented | 501 Not Implemented | The request method is not supported by the server and cannot be handled. |
| BadGateway | 502 Bad Gateway | The server received an invalid response from the upstream server. |
| ServiceUnavailable | 503 Service Unavailable | The server is not ready to handle the request, often due to maintenance. |
| GatewayTimeout | 504 Gateway Timeout | The server did not receive a timely response from an upstream server. |

#### Custom Error Schema

Error responses conform to the following JSON schema:

```
{
  "data": {},
  "items": [
    {
      "code": "string",
      "message": "string",
      "severity": "enum (Possible values: \"NotSet\", \"Information\", \"Warning\", \"Error\")",
      "classification": "enum (Possible values: \"Business\", \"System\")"
    }
  ]
}
```

#### Request Payload Storage in Application Insights

- When an exception occurs, the **request payload** is **encrypted** and stored in **Application Insights** as a property within the custom dimensions.
- Sensitive values are securely stored in **Azure Key Vault**.

#### Handling Custom Correlation Id ( `X-Correlation-Request` )

- If a custom correlation ID is **not present** in the HTTP request header, the exception handling middleware **generates one** and assigns it to the header.
- If a correlation ID **already exists**, the middleware logs the exception in Application Insights using the **existing correlation ID**.

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#### API Performance and Capacity Metrics

| Metric Type | Metric | Value |
|---|---|---|
| **Functional Load** | Error rate<br>[Failed requests percentage per minute] | 0.01% |
| | Availability | 100% |

| | | |
|---|---|---|
| | [System uptime percentage] | |
| **Latency** | Average response time | 1 sec |
| | 90th percentile | 1.3 sec |
| | Worst case | 1.5 sec |
| Throughput | Average requests per minute | 800 calls/min |
| | Peak requests per minute | 2100 calls/min |
| Expected Resource Use | Limits: Memory | 1Gi |
| | Requests: CPU | 100m |
| | Requests: Memory | 750Mi |
| | HPA Configuration: Min Replicas | 2 |
| | HPA Configuration: Max Replicas | 10 |

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### Infrastructure and Deployment Design

At the API level, define

- AKS namespace and container info:
  - Namespace: **hicbocoreservices-{env}-ns**
  - Container: **hi-api-sync**
- Ingress/APIM setup
  - APIM API Name - **hisyncdomain/{env}**
- ConfigMap and Secret usage
  - Library Variable Group: **hicbocoreservices-sync-api-{env}-vars**
- Infrastructure CI/CD pipeline details:
  - **<To be Provided later>**
- Environment strategy (Dev, T27, Prod, etc.)
  - DEV: **d24**, SIT: **d34**, PERF: **t33**, UAT:**euat**, PROD: **prod**

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### Observability

#### AppDynamics/App Insights Integration Setup

- AppDynamics Tier/node mapping: NA
- AppDynamics Custom business transactions and metrics: NA
- App Insights: **hicbocoreservices-{env}-ai01**

#### AppDynamics/AppInsights Alerts

- CPU/memory/error rate thresholds: Api Alerts
- Notification channels: ServiceNow Ticket and Email Alert to dotnet support team.

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### Repository and Pipelines

| Details | Link |
|---|---|
| Git repository | https://bupaaunz.visualstudio.com/HI-CBO-Apps/_git/HI.API.Sync |
| Branching strategy | Branching Strategy for .NET based projects - DotNet Team - Confluence |
| Build and release pipelines | To be updated later |
| Security scans, tests, validations | To be updated later |

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### Non-Functional Requirements

- Scalability: HPA should be enabled.
- Resilience and failover strategy (DR): API will be Active - Active running in both AKS clusters **productionaks** and **productiondraks.** For infrastructure DR refer the document:
  - Hugo Core Services and Batch Jobs Technical Recovery Procedure - Disaster Recovery
- Security and compliance standards (e.g., OWASP): SCR_008 API Security & Management
- Rate limiting and throttling: Rate limited on the API Level to **2100 requests per IP/min**

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### Secrets Management

- Use of Key Vault or secret manager: Key Vaut will be used **banz-hi-sync-{env}-kv01**
- Rotation and access control:
  - Manual yearly rotation of key vault entries has to be done for the following **<More details will be filled after the initial release>**

| Key Vault Value | Description |
|---|---|
| ApimKeyValue | Subscription key for authenticating with Azure API Management (APIM). |

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#### API Specification

The below link has detailed design per end point

Sync